Sebastian Ljung

https://sebastianl2014.github.io sl2014@cam.ac.uk

Education

University of Cambridge, Peterhouse

- Natural Sciences, doing 4th year integrated masters in Astrophysics. ~11 months internship experience in embryology (statistical analysis and clustering) and astrophysics.
- **Research review** (1st class), Kavli Insitute for Cosmology [4 months, part-time]: 2022
  ↳ On first galaxies with JWST, with Dr. Laporte (Prof. Tacchella's group). >28 months of group meetings (still going)
- **Computing project**[AI] (1st class) [4 weeks]: Vorticies in Incompressible Schrödinger Flow. 2024
  ↳ Implemented model from a paper to investigate interplay between vortex element evolution & plank length.
  Used $\mathbb{C}^2$ field solutions to a modified Schrödinger eq. as hydrodynamic simulation.
- MSc: Advanced Stellar Evolution, Astro-statistics, Rel. astrophys. & cosmology (+lectures: GR, QFT)
- BSc: General relativity, Astro-fluid dynamics, Electrodynamics & Optics, Advanced QM, Stat-mech, Classical field theory & phase transitions, Particle & nuclear physics, Scattering theory.

Publications

- Taubenschmid-Stowers J., Rostovskaya M., Santos F., Ljung S., ..., Reik Wolf 2022
  "Modelling human zygotic genome activation in 8C-like cells in vitro"

Research Experience

- **MSc project supervised by Prof. Tacchella & Dr. Gilkis, Kavli Institute of Cosmology:**
  ↳ Part of collaboration, (modifying MESA stars code, in Fortran):
  → (For Stars group): Implement star-disk model to enable stable modelling of differentially rotating binaries.
  → (For Extragalactic group): Then generate grid of binaries, used for spectral fitting of high-z galaxies.
  - Independently formulated analytic solution to Prof. Paczynski's (1991) numerical model on polytropic accretion disks around critically rotating main sequence (MS) stars.
  - Incorperated supernovae & evolution of compact object binaries within MESA.
- **Internship in Prof. Anastasia Fialkov's group, Institute of Astronomy [8 weeks]:** 2023
  ↳ Modified semi-analytic A-SLOTH simulation (Fortran) to account for the influence of super-massive black holes (SMBH) on the modelled nuclear stellar clusters & Pop III black hole binaries.
  - After literature review, proposed a side-project to introduce SMBHs
  - Incorporated equations from publications and justified modelling approximations.
  - Implemented and argued for significance of eccentricity on Stochastic GW Background.
  - Prof. Roman Rafikov recently (Oct) offered to help guide this to publication, after MSc.
- **Internship at Dr Michael Imbeault lab, Genetics Dept. Cambridge [10 weeks]:** 2022
  - Handling large datasets (>30GB), w/o cluster, in a RAM and time-efficient manner.
  - Spearheaded side-project, and liaised with Prof. Reik's, lab, to determine potential ZGA targets for collaboration, using developed tool.
- **Internship at Prof. Wolf Reik lab, BI Cambridge [9 weeks]:** 2021
  ↳ Suggested & applied tool that uses 1st-order ODEs & dimensionality reduction to predict cell state transitions. (analogous to phase-space flow in galactic dynamics.)
  - Co-authorship for answering review question, computationally bypassing months of costly experiments.
  - Presented results of experimental + computational (side-project) at conference poster session.

(These are my own work and made limited use of ChatGPT/AI. Exceptions: Grammarly spell-checks, conceptual code explanations, syntax errors, & projects where AI use was aknowled